6. Write a Servlet to demonstrate session tracking using HttpSession. Implement a simple login system where the user's session is tracked.

AIM

To create a web application that demonstrates session tracking using the `HttpSession` interface. The application will maintain a user's login state across multiple pages of the Department Website (AI & ML)

ALGORITHM

1. **Input:** The user enters their credentials on the login page.
2. **Session Creation:**
   - Upon successful authentication, the Servlet creates a session using `request.getSession()`.
   - The user's identity is stored as a session attribute (e.g., `session.setAttribute("user", name)`).
3. **Session Tracking:**
   - When the user navigates to the AI-ML Dashboard, the Servlet checks if a valid session exists.
   - If `session.getAttribute("user")` is not null, the dashboard is displayed; otherwise, the user is redirected to the login page.
4. **Session Termination:** A "Logout" button is provided to invalidate the session using `session.invalidate()`, ensuring the user is logged out.
5. **Deployment:** Compile the Servlet, map it in the `web.xml` or via annotations, and deploy it to Apache Tomcat.

SOURCE CODE

A. HTML File (index.html)

This file provides the login interface for the weather portal.

HTML
```
<!DOCTYPE html>
<html>
<head>
<title>Login</title>
<style>
body { font-family: Arial; text-align: center; margin-top: 100px; }
input, button { padding: 10px; margin: 5px; }
#dashboard { display: none; }
</style>
</head>

<body>

<!-- Login -->
<div id="login">
  <h2>Login</h2>
  <input type="text" id="user" placeholder="Username"><br>
  <input type="password" id="pass" placeholder="Password"><br>
  <button onclick="login()">Login</button>
  <p id="msg"></p>
</div>
```

```
<!-- Dashboard -->
<div id="dashboard">
  <h2>Welcome <span id="name"></span></h2>
  <p>Session Active</p>
  <button onclick="logout()">Logout</button>
</div>

<script>
let session = {};

function login() {
  let u = user.value;
  let p = pass.value;

  if (p === "admin123") {
    session.user = u;
    login.style.display = "none";
    dashboard.style.display = "block";
    name.textContent = u;
  } else {
    msg.textContent = "Wrong Password";
  }
}

function logout() {
  session = {};
  dashboard.style.display = "none";
  login.style.display = "block";
}
</script>

</body>
</html>
```

Java File (SessionServlet.java)

This Servlet manages the session creation and the weather dashboard.

Java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if (password.equals("admin123")) {
            HttpSession session = request.getSession();
            session.setAttribute("user", username);
            response.sendRedirect("dashboard");

        } else {
            out.println("<h3 style='color:red;'>Invalid Password</h3>");
        }
```

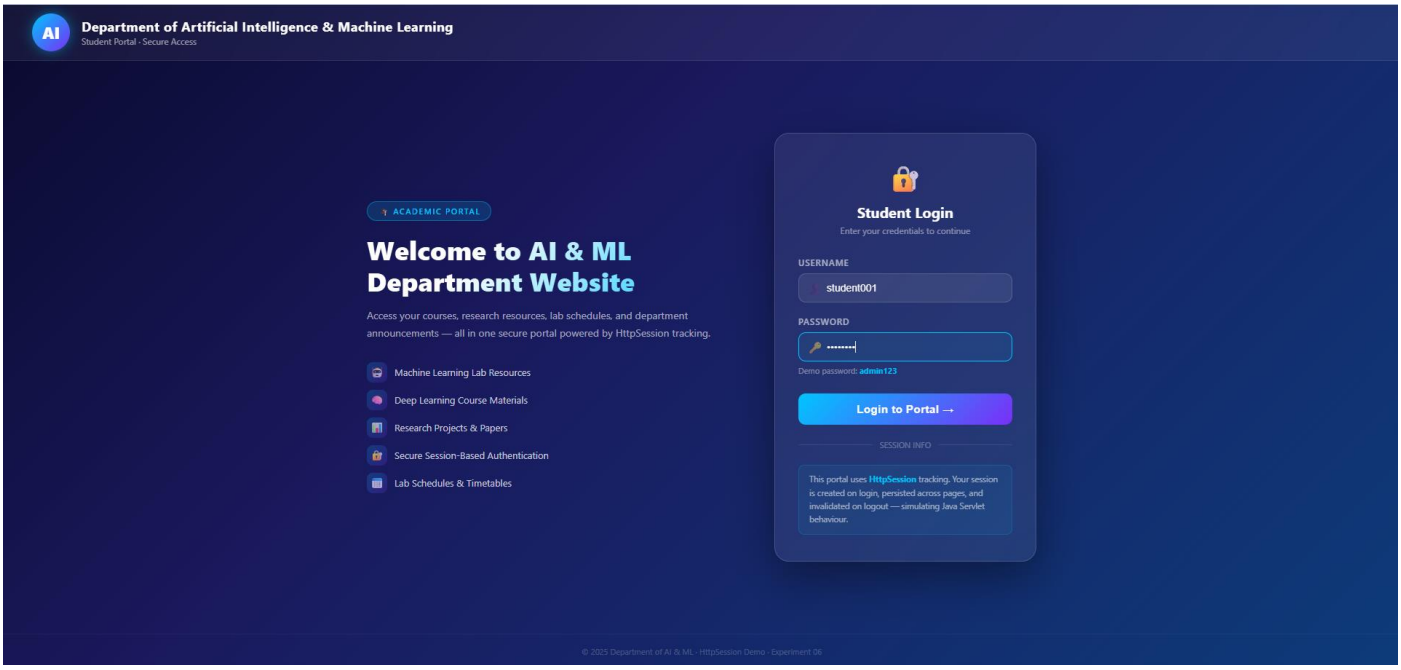
```
    }
}
```

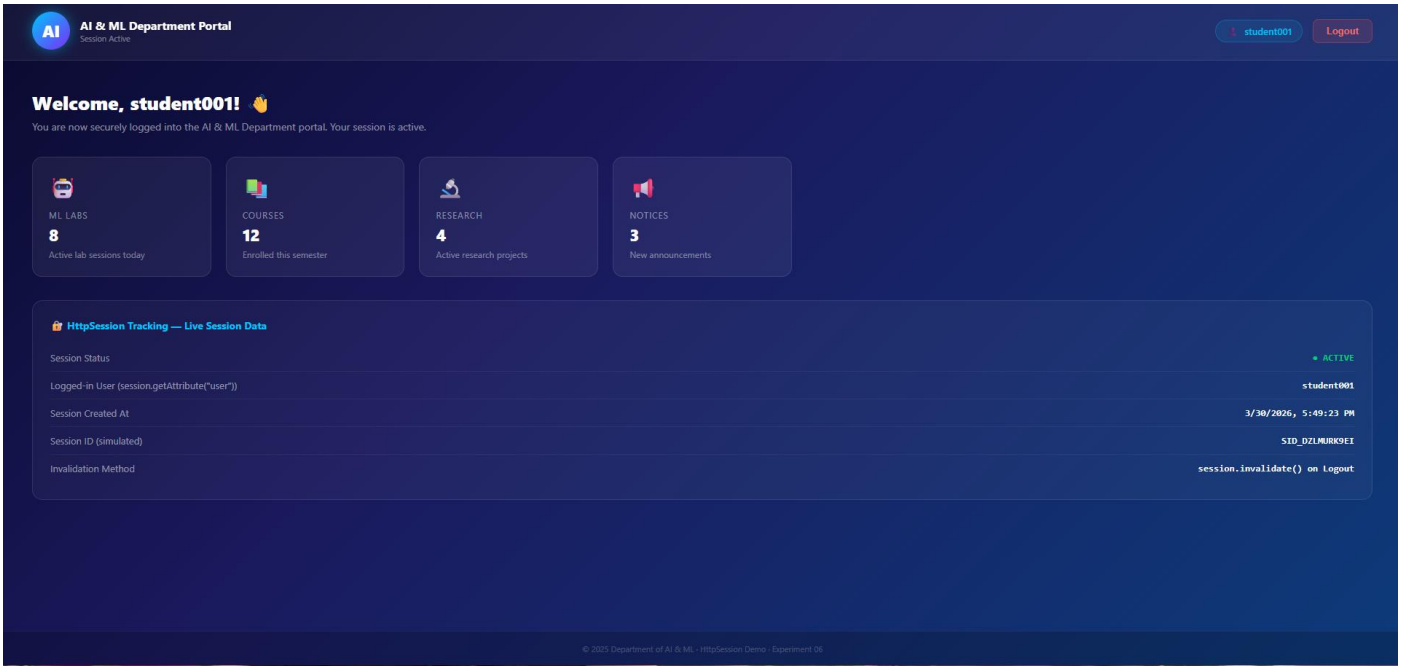
Java File (DashboardServlet.java)

This Servlet verifies the session before showing weather data.

Java
```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/dashboard")
public class DashboardServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession(false);

        if (session != null && session.getAttribute("user") != null) {

            String user = (String) session.getAttribute("user");

            out.println("<html><head><title>Dashboard</title></head><body>");
            out.println("<h2>Welcome " + user + "!</h2>");
            out.println("<p>Session is ACTIVE</p>");

            // Session details
            out.println("<p>Session ID: " + session.getId() + "</p>");
            out.println("<p>Created Time: " + new
java.util.Date(session.getCreationTime()) + "</p>");

            out.println("<br><a href='logout'>Logout</a>");
            out.println("</body></html>");

        } else {
            // If no session, redirect to login page
            response.sendRedirect("login.html");
        }
    }
}
```

OUTPUT:


AI
Department of Artificial Intelligence & Machine Learning
Student Portal · Secure Access
ACADEMIC PORTAL
Welcome to AI & ML Department Website
Access your courses, research resources, lab schedules, and department announcements — all in one secure portal powered by HttpSession tracking.
Machine Learning Lab Resources
Deep Learning Course Materials
Research Projects & Papers
Secure Session-Based Authentication
Lab Schedules & Timetables
Student Login
Enter your credentials to continue
USERNAME
student001
PASSWORD
Demo password: admin123
Login to Portal →
SESSION INFO
This portal uses HttpSession tracking. Your session is created on login, persisted across pages, and invalidated on logout — simulating Java Servlet behaviour.
© 2025 Department of AI & ML · HttpSession Demo · Experiment 06


AI
AI & ML Department Portal
Session Active
student001
Logout
Welcome, student001! 👋
You are now securely logged into the AI & ML Department portal. Your session is active.
ML LABS
8
Active lab sessions today
COURSES
12
Enrolled this semester
RESEARCH
4
Active research projects
NOTICES
3
New announcements
HttpSession Tracking — Live Session Data
Session Status
ACTIVE
Logged-in User (session.getAttribute("user"))
student001
Session Created At
3/30/2026, 5:49:23 PM
Session ID (simulated)
SID_DZLMURK9EI
Invalidation Method
session.invalidate() on Logout
© 2025 Department of AI & ML · HttpSession Demo · Experiment 06

RESULT

A Java Servlet application was successfully implemented to demonstrate session tracking using `HttpSession`. The application effectively maintained user data across different web pages within the Weather Information portal.